

Analysis Quality Score Explanation for ResearchLens

This document explains how ResearchLens computes the evaluation metrics shown in the dashboard:

- Topic Coherence
- Topic Coverage
- Gap Novelty
- Model Quality

It also explains the additional summary values such as high-confidence gaps and average papers per topic.

1. Metric definitions

Topic Coherence

- Computed as the average coherence score of all detected topics.
- Each topic already includes a **coherence** value from the topic modeling step.
- Formula:

```
topicCoherence = sum(topic.coherence for each topic) / number_of_topics
```

- Displayed as a percentage by multiplying the result by 100.
- Example: **0.64** becomes **64%**.

Topic Coverage

- Measures how many papers were assigned to at least one topic.
- The UI computes it as the fraction of unique assigned papers over the total number of uploaded papers.
- Formula:

```
topicCoverage = assignedPaperCount / totalPaperCount
```

- In another summary view, it is also computed as the total number of papers referenced by topics divided by the total papers, capped at 1.
- Example: **1.00** becomes **100%**, meaning every paper was assigned to a topic.

Gap Novelty

- Computed as the average **gapScore** across all detected gaps.
- The score is capped at 1 to remain within a standard percentage range.
- Formula:

```
gapNovelty = min(1, sum(gap.gapScore for each gap) / number_of_gaps)
```

- Displayed as a percentage. Example: **0.16** becomes **16%**.

Model Quality

- A composite quality metric that combines coherence, coverage, and gap novelty.
- The evaluation section uses this formula:

```
modelQuality = round((avgCoherence * 0.4) + (topicCoverage * 0.3) + (gapNovelty * 0.3), 2)
```

- This means:
 - Topic Coherence has the largest influence (40%).
 - Topic Coverage and Gap Novelty each contribute 30%.
- Example: with 64% coherence, 100% coverage, and 16% gap novelty, the model quality is roughly **0.60** or **60%**.

2. High-confidence gaps

- The dashboard counts gaps with a **gapScore** greater than **0.5**.
- Formula:

```
highConfidenceGaps = number of gaps where gap.gapScore > 0.5
```

- In the summary line, this is reported as **0 high-confidence gaps detected (score > 0.5)** when no gap passes that threshold.

3. Processing summary values

Processing completed in live

- The UI shows **live** when the current run is being displayed in real time rather than using an archived timing value.

Average papers per topic

- Computed from the total paper count divided by the number of detected topics.
- Formula:

```
avgPapersPerTopic = totalPapers / numberOfTopics
```

- Example: **4.4 papers per topic** means the dataset was split into topics such that each topic contains an average of 4.4 papers.

4. Interpreting the example scores

- **Topic Coherence 64%**: topics are moderately coherent; papers within each topic are reasonably aligned.
- **Topic Coverage 100%**: all papers were grouped into topics, so there are no orphan papers.
- **Gap Novelty 16%**: the detected gaps are relatively low in novelty, meaning the topic intersections are not highly surprising.
- **Model Quality 60%**: the overall evaluation is pulled down by the low gap novelty, despite full coverage and moderate coherence.

5. Why these metrics matter

- **Topic Coherence** helps you understand whether topics are meaningful and focused.
- **Topic Coverage** checks if the system successfully placed all papers into the topic model.
- **Gap Novelty** suggests how valuable the discovered research gaps are for new directions.
- **Model Quality** summarizes all three metrics into a single quality signal for the analysis pipeline.